QUALIFICATION STANDARD
AND PLAN (QSP)

CRYPTO KY-99M Modern Secure Radio
Voice and Data Encryptor Operator (KY-99M Operator)
NQual (ALYX)

MITE ID
Residential Phase – 122216

Regular Force: Naval Communicator 00299- 01 (Nav Comm)
Primary Reserve: Naval Communicator 00299-04 (Nav Comm)
Training Authority (TA): Commander Naval Training Group (CNTG)
Curriculum Control Authority (CCA): NPTG NTDC(A)

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RECORD OF AMENDMENTS

1. The following versions of this QSP have been approved for use in support of the conduct of training. Amendments detailed in Annexes D and/or L.

Revision	Change Description	Date	Authority
1.0 (QSWB Month Year)			
			NTG HQ
1.01			CCA

NOTE:

Personnel programming courses must ensure that their hard copy of this document is current and identical to the electronic copy on the NTG Website. See above for current changes.

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FOREWORD

1. This QSP is issued on the authority of the Commander of the Royal Canadian Navy and managed by the Commander Naval Personnel and Training Group (CNPTG) on behalf of the Assistant Chief of Naval Staff Personnel and Training (A/CNS P&T).

2. This QSP is based upon the Specification Type, Specification Name published Day Month Year A-PD-055-002/PP-001;

3. This QSP publication is effective on receipt. This QSP is designed to assist in developing, conducting, evaluating, and validating the Individual Training and Education (IT&E) required for training.

4. The QS at Chapters 1 and 2 were developed by the Qualification Standard Writing Board (QSWB) convened 08 -11 August 2023 at CFB Esquimalt working in collaboration with subject matter experts (SMEs) drawn from participating organizations. The subsequent TP at Chapters 3 and 4 were developed by the Training Plan Writing Board (TPWB) convened 3-13 Oct 2023 at CFB Halifax working in collaboration with subject matter experts drawn from participating organizations. All aspects of this document were prepared in accordance with the concept of training outlined in A-P9-050 Series, Canadian Forces Manual of Individual Training and Education. Full details of the WBs are contained within Annex D and L.

5. The criteria for major and minor amendments, and the approval process, are identified in NPTORD 4500-22.

6. In circumstances where the QS and/or TP cannot be adhered to, submission of a QSP Deviation Request is required.

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CHAPTER 1 QUALIFICATION STANDARD

AIM

1. The aim of this QSP is to establish the requirements and training for CAF / DND personnel employed in settings where secure communications are required utilizing the KY-99M Modern Type 1 Encryptor Device (KY-99M). Possession of this qualification will allow personnel to configure the KY-99M to support encrypted voice and data communications.

GRANTING OF NQUAL

2. A trainee will achieve the qualification ALYX when the standards listed in the Performance Objectives (POs) have been met.

CO-QUALIFICATION

3. There are no co-qualifications for this course.

PREREQUISITES

4. To be selected for this course the loading authority must ensure that trainees meet the following prerequisites:

   a. Communications personnel who are employed or selected to soon to be employed in a position that installs, configures, and maintains KY-99M;

   b. Must have attained OFP in their respective MOSID or equivalent for DND;

   c. Have a minimum security clearance level of Enhanced Reliability.

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CHAPTER 2 PERFORMANCE OBJECTIVES

PO 001 CONFIGURE THE KY-99M FOR COMMUNICATION

1. Performance: Configure KY-99M for communication

2. Conditions:

   a. Given:

      (1) KY-99M Secure Radio Data and Voice Encryptor;
      (2) KY-99M ancillary equipment;
      (3) OpTask Comm;
      (4) Simple Key Loader (SKL) preloaded with keymat;
      (5) Cryptographic Ignition Key (CIK) (KOV-80);
      (6) Job Aids; and
      (7) References.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Any communications environment.

3. Standard: The communications personnel shall (XT0001) configure KY-99M for communication IAW A1 and in compliance with CAF/DND security policy and procedures by:

   a. (XT0005) Determining circuit requirements for operation from the OpTask Comm to include:

      (1) Encryption requirements of the circuit

      (2) Frequency band required

      (3) Keymat short title

   b. (XT0007) Verifying KY-99M equipment status IAW A1 Section 4 & 8 by:

      (1) Ensuring CIK is installed and fully seated;

      (2) Ensuring power to KY-99M;

      (3) Identifying built-in test (BIT) result;

      (4) Implementing troubleshooting procedures IAW A1 to address faults;

   c. (XT0006) Verifying configuration of KY-99M by:

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(1) Selecting a preset IAW circuit requirement; and

(2) Confirming the selected preset defaults IAW Online Configuration Sheet.

d. (XT0002) Loading keymat into KY-99M equipment IAW A1 Section 4.

e. (XT0008) Confirming end-to-end functionality of the circuit via radio check by:

(1) Ensuring the KY-99M mode switch is set to the online mode (see remark para c) IAW with circuit requirements;

(2) Establishing end-to-end communication via associated connected communications equipment; and

(3) Informing the chain of command (CoC) that circuit is set up with KY-99M.

f. (XT0009) Conducting KY-99M shutdown procedure as directed by Coc by:

(1) (XT0003) Zeroizing KY-99M keymat associated to the circuit(s) IAW A1 Section 4; and

(2) Powering off KY-99M IAW A1 Section 3

4. References: A1, job aids, DND Infosec / Comsec Policy

5. Tasks: XT0001, XT0002, XT0003, XT0005, XT0006, XT0007, XT0008, XT0009

6. Remarks:

a. Note to TPWB for training purposes the OpTask Comm and keymat shall be unclassified material

b. Ancillary equipment includes cabling (power / network), standalone laptop with Windows-based operating system, PTT handset, and power source

c. KOV-80 CIK is specific to KY-99M equipment

d. Job Aids include:

(1) Online Configuration Sheet; and

(2) Offline Configuration Sheet

e. In para 3e1 of this PO, online mode refers to plain text (PT), cypher text (CT) 1 through 6, and manual modes, when operating the KY-99M.

f. Knowledge, Skills and Attitudes are identified in the Task Disposition at Annex C and shall be considered during the design of the training during the TPWB.

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TRAINING PLAN

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CHAPTER 3 TRAINING MANAGEMENT DETAILS

TRAINING STRATEGY

1. The training strategy selected for this course is residential.

TRAINING DURATION

2. Based on existing resources and approved instructional design requirements, the approved Course Design authorizes 3 training days.

REQUIRED COURSE STAFFING

3. The following key personnel are required to conduct the training.

Course Director				
Rank	MOSID	Qualifications Required	Qty	PO / EO
PO1	00299	QL6	1	ALL
Instructors				
Rank	MOSID	Qualifications Required	Qty	PO / EO
S1	00299	QL5A / RQ S1 / KY-99M Operator	1	ALL

Guest Lecturers/Others (May be civilian)				
Lecturer	# Instructor	Subject	EO #	# period
N/A				

TRAINING LIMITATION

4. A training limitation may be due to constraints of resources, personnel or equipment. A training limitation is the difference between what a qualified individual must be able to do and what training can prepare them to do. There is no training limitation.

COURSE CAPACITY

5. The capacity per session is based on the guidelines in CFITES Vol 5 – Instructor to Learner Ratios:

   a. Optimum – 8 trainees;

   b. Maximum Load –12 trainees; and

   c. Minimum – 4 trainees

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DESIGNATED SCHEDULING AND COURSE LOADING

6. Training will be scheduled utilizing the HRMS Training (MITE) module to meet the training requirements (IAW NPTORD 4540-2).

7. The authority to load members onto this course will be IAW NPTORD 4540-43. NTG HQ is to be included as an info addressee on all loading messages. Course loading will be accomplished by:

a. The Career Manager for career courses; or

b. NFS Campus Support for occupation and specialty courses.

QUALIFICATION AND ASSESSMENT

8. This is a pass/fail course. Students must successfully complete all POs to achieve the qualification.

9. Students will be monitored and assessed both on performance and conduct throughout the course. Counselling, progress monitoring, and maintenance of student files will be in accordance with NPTORD 4500-54. Although not mandated specifically in the POs, the general demands of service members and the leadership demands of junior and senior leaders form an integral part of individual training. As such, students are presumed to possess the leader and/or follower attributes commensurate with their rank.

UNSATISFACTORY PROGRESS

10. Any student who fails to successfully complete all POs in the allocated number of attempts will be referred to the Commandant (Cmdt) or Commanding Officer (CO) of the training establishment (TE) to determine the requirement for a TRB IAW NPTORD 4500-54.

REMOVAL FROM TRAINING

11. A TE CO/Cmdt may direct that a student be removed from training for reasons of deficiency in performance, conduct (discipline, attitude, teamwork, safety, leadership), missed training, medical, and other non-performance/conduct related issues. In these cases, Cease Training Procedures are to be strictly adhered to IAW NPTORD 4500-29 or 4500-54.

RECERTIFICATION REQUIREMENTS

12. Nil.

LANGUAGE OF INSTRUCTION

13. The language of instruction shall be in accordance with the Delivery of Training and Education in Both Official Languages (DAOD 5039-6).

PRIOR LEARNING ASSESSMENT AND RECOGNITION (PLAR)

14. PLARs shall be conducted in accordance with DAOD 5031-1 Canadian Forces Military Equivalencies Program, through the RCN PLAR submissions coordinator.

POST COURSE REVIEWS (PCR)

15. PCRs are to be conducted on all steady state courses IAW NPTORD 4500-23.

COURSE REPORTING

16. Formal Course: A Canadian Forces Course Report, Form CF 377 and/or CF289 Certificate of Military Achievement, shall be prepared and distributed in accordance with DAOD 5031-9. The Course Report shall be descriptive, identifying the student's strengths and weaknesses with specific comments on any PO for which the student demonstrated outstanding performance or had difficulty achieving the standard, any conduct relevant to course performance, and any conditions or situations that affected course performance.

CHAPTER 4 LESSON SPECIFICATIONS

PO 001 – CONFIGURE THE KY-99M FOR COMMUNICATION

EO 001.01 Initialize KY-99M for operational use

1. Performance: Initialize KY-99M for operational use

1. Conditions:

    a. Given:

        (1) KY-99M;

        (2) Laptop Windows operating system;

        (3) Ancillary communication equipment;

        (4) CIK;

        (5) References;

        (6) Assistance; and

        (7) Scenario with directions;

    b. Denied: Assistance during PC;

    c. Environment:

        (1) Classified classroom; and

        (2) Day or night.

2. Standard: The trainee shall initialize KY-99M for operational use by:

    a. (XT0007) Verifying KY-99M equipment status IAW A1 Sect 4 & 8 by:

        (1) Ensuring the matching CIK is installed and fully seated by:

            (a) Verifying CIK serial number match with KY-99M; and

            (b) Inserting CIK into receptacle until security pin raises.

        (2) Ensuring power to KY-99M by turning selector/mode switch to “PT”;

    b. Implementing troubleshooting procedures IAW A1 Sect 8.5 to include:

        (1) Recognizing CIK error messages;

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(2) Identifying built-in test (BIT) results; and

(3) Addressing faults.

c. (XT0005) determining circuit requirements for operation from OpTask Comm to include:

(1) Encryption requirements of the circuit;

(2) Keymat short title; and

(3) Frequency band required;

d. (XT0006) Verifying configuration of KY-99M to include:

(1) Configure ECU settings with End Cryptographic Unit Interface (ECUI) IAW A1 Sect 4.2.5. by:

(a) Turning Selector/mode switch to “OFL”;

(b) Selecting a preset IAW with circuit requirements and as directed; and

(c) Confirming the selected preset defaults IAW A1 sect 10.

(2) Configure ECU settings with Operational and Configuration Management Interface (OCMI) by:

(a) Enabling OCMI through ECUI IAW A1 Sect 4.2.5.;

(b) Turning Selector/Mode Switch to “remote”;

(c) Connecting laptop to OCMI port;

(d) Modifying NIC set up to allow connection through OCMI port IAW A1 Sect 11.1;

(e) Logging in to maintainer account through Web Browser; and

(f) Configuring settings IAW A1 Sect 11 and operational requirements;

e. (XT0009) Conducting KY-99M shutdown procedure as directed by Coc by powering off KY-99M IAW A1 Sect 3.

3. Teaching Points:

Teaching Points	Instructional Strategies	Time	References/Remarks
1. Describe the KY-99M to include: - Capabilities and features. - Functions - Firmware and Personalities - Interoperable Equipment - Key Management - Specifications. - Accessories	Method: Interactive Lecture Media: Power Point/Real Equipment Environment: Classroom	1x50 min	A1 Sect 1 and 2
2. Demonstrate insertion of CIK by: - Identification of CIK Serial number - Physical Insertion into CIK Receptacle			Need to be able to identify the correct CIK before insertion A1 Sect 2.5
3. Conduct start up/shut down procedures			A1 Sect 2.4
4. Recognize CIK error messages to include: - Invalid CIK - Missing CIK			A1 Sect 8.5
5. Identify BIT results			Problem solving guide required A1 Sect 7.5 and 8.5
6. Extract operationally relevant data to operate the KY-99M from the OpTask Comms	Method: Interactive Lecture Media: Power Point/Real Equipment Environment: Classroom	1x50 min	Identifying which type of circuit and understanding what the use of that circuit is crucial to patching it
7. Define Wideband and Narrowband to include: - Explain the difference between Wideband and Narrowband - Explain characteristics of each - Explain functions of each - Identify proper channels on KY-99M (under factory defaults)			

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Teaching Points	Instructional Strategies	Time	References/Remarks
8. Manipulate ECUI (Off-Line KY-99M Menus): - PRESETS - INTERFACE OPS - SET DEFAULTS - MENU LOCK - TEST OPS and - KEY OPS	Method: EDI Media: Power Point/Real Equipment Environment: Classroom	1x50 min	A1 Sect 3 and 4
9. Configure ECU through OCMI to include: - Enabling OCMI through ECU IAW A1 sect 4.2.5; - Connecting laptop to OCMI port - Configuring settings IAW A1 sect 11 and operational requirements.	Method: EDI Media: Power Point/Real Equipment Environment: Classroom	2x50 min	A1 sect 4.2.5.; A1 sect 11
10. Modify network settings to have appropriate IP address for the OCMI			A1 Sect 11.1
11. Differentiate between operator and maintainer functions in OCMI			A1 Sect 11.2
12. Select a preset channel and confirm the configuration status.			Either way of confirming the config status would be acceptable A1 Sect 4.1
13. Identify troubleshooting requirements to include: - LCD screen unreadable - Wrong IP address in OCMI			A1 Sect 8.5

4. Total EO Time: Total: 5 x 50 minute periods as follows:

   a. 5 x 50 min period; and

   b. 0 x 50 min periods – EC

5. Training and Learning Aids: Training OpTask Comms

6. Remarks: Nil.

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EO 001.02 Load keymat into KY-99M equipment

1. Performance: Load keymat into KY-99M equipment

2. Conditions:

   a. Given:

      (1) KY-99M;

      (2) Laptop Windows operating system;

      (3) Ancillary communication equipment;

      (4) CIK;

      (5) SKL;

      (6) Training Cryptographic keymat;

      (7) References;

      (8) Assistance; and

      (9) Scenario with directions;

   b. Denied: Assistance during PC;

   c. Environment:

      (1) Classified classroom; and

      (2) Day or night.

3. Standard: The trainee shall (XT0002) Load keymat into KY-99M equipment IAW A1 Sect 4 to include:

   a. Transfer Keymat from Simple Key Loader (SKL) or Really Simple Key Loader (RASKL) to KY-99M IAW operational requirements and A1 Sect 4.7 to include;

      (1) Profile load; and

      (2) Stray load;

   b. Zeroizing KY-99M keymat associated to the circuit(s) when needed IAW A1 Sect 4 by:

      (1) if single key, by:

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(a) Turning selector/mode switch to “OFL” IAW A1 Sect 4; and

(b) Following single key zeroizing procedures IAW A1 Sect 4.6;

(2) If All keys, by:

(a) Turning selector/mode switch all the way clockwise pulling out to “Z all”

(b) Monitoring ECU for power cycle instructions; and

(c) Cycling power;

(3) Confirming zeroization procedures by entering Key Ops menu to view any remaining keys;

c. (XT0008) Confirming end-to-end functionality of the circuit via radio check by:

(1) Ensuring the KY-99M selector/mode switch is set to the online mode IAW circuit requirements;

(2) Establishing end-to-end communication via associated connected communications equipment until end-to-end communication is successful or troubleshooting is required;

(3) Implementing troubleshooting procedures IAW A1 Sect. 8.5 by:

(a) Recognizing the appropriate source of the error; and

(b) Addressing faults;

(4) Informing the CoC that circuit is set up with KY-99M as directed.

4. Teaching Points:

Teaching Points	Instructional Strategies	Time	References/Remarks
1. Differentiate PT and CT: - Explain their use - identify display message	Method: Guided Discussion Media: PPT Environment: Classroom	1 x50 min	Instructor is to explain and address faults as they happen through EDI. A1 Sect 7

Teaching Points	Instructional Strategies	Time	References/Remarks
2. Differentiate Stray vs. Profile load: - When to use each - Which load type is applicable for the situation.			A1 Sect 4.7
3. Build SKL Equipment Profile for KY-99M			
4. Use SKL to transfer Keymat to KY-99M to include: - Profile load; and - Stray load	Method: EDI Media: Power Point/Real Equipment Environment: Classroom	1x50 min	Instructor is to explain and address faults as they happen through EDI. A1 Sect 4.7
5. Zeroize single Key following zeroizing procedures IAW A1 Sect 4.6	Method: EDI Media: Power Point/Real Equipment Environment: Classroom	1x50 min	Instructor is to explain and address faults as they happen through EDI. A1 Sect 7.5
6. Zeroize all keys by: - Turning selector/mode switch all the way clockwise pulling out to "Z all" - Monitoring ECU for power cycle instructions - Cycling power.			
7. Confirm zeroization procedures by entering Key Ops menu to view any remaining keys			
8. Ensuring the KY-99M selector/mode switch is set to the online mode IAW circuit requirements;	Method: EDI Media: Power Point/Real Equipment Environment: Classroom	1x50 min	Instructor is to explain and address faults as they happen through EDI. A1 Sect 5
9. Establish end-to-end communication via associated connected communications equipment until end-to-end communication is successful or troubleshooting is required			

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Teaching Points	Instructional Strategies	Time	References/Remarks
10. Recognize and address potential faults to include: - Wrong keymat loaded - Superseded keymat - Incorrect/broken physical components - Incorrect profile build - Attempting comms on different channels during radio check - Mis-matched settings	Method: Guided Discussion Media: Instructor Environment: Classroom	1x50 min	
11. Communicate verbally with appropriate authority to include: - What troubleshooting steps have been taken if circuit setup unsuccessful - Inform CoC if successful circuit setup			
12. Loading keymat in KY-99M	Method: Practical lab Media: Instructor Environment: Classroom	2x 50 min	A1 Labs from the manufacturer
13. Configure KY-99M for communication	Method: Practical lab Media: Instructor Environment: Classroom	2x 50 min	A1 Labs from the manufacturer

5. Total EO Time: Total: 9 x 50 minute periods as follows:

   a. 9 x 50 min period; and

   b. 3 x 50 min periods – EC (150 min).

   c. 2 x 50 min periods – PC (100 min).

   d. 2 x 50 min periods – re-test PC (100 min).

6. Training and Learning Aids:

   a. Practical Labs; and

   b. Scenarios

7. Remarks: Practical Labs are to be found with the ICT instructors, so we can reuse them.

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QUALIFICATION STANDARD ANNEXES

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ANNEX A – MAIN REFERENCES

1. This annex lists all references used in this QS and TP.

REF #	NDID/DOCUMENT CONTROL NO	TITLE OF REFERENCE
A		**Official CAF Publications and Directives that relate to this Qualification**
A1		KY-99M Modern Secure Radio Voice and Data Encryptor Version 9.1.2
A2		
A3		
A4		
B		**(Non-CAF) Government of Canada Publications & Foreign Military (Non-DND) Publications that relate to this Qualification**
B1		
B2		
B3		
C		**Civilian Publications and Manuals that relate to this Qualification**
C1		
C2		
D		**Other Publications and Manuals that relate to this Qualification**
D1		
D2		

NOTE: If additional references are identified during the TPWB this annex can be updated secretarially through NTG HQ.

ANNEX B – TASK LIST

1. The task list includes all of the tasks associated with this qualification, and the outcome of the Train / No-train decisions. If a task is designated as requiring training, the PO where it is covered is indicated. If a task is designated as not requiring training, the justification for that designation is provided.

2. The following table defines some of the criteria used to support a no-train designation in the task list. Other reasons may be used if required.

CRITERIA	DEFINITION
A	**Previously qualified.** Tasks and knowledge already possessed by the student by CAF and/or RCN courses, training and/or experience.
B	**Will receive on future courses.** Tasks and knowledge will be covered as part of other CAF and/or RCN courses and/or training.
C	**Difficulty.** Task is not difficult to learn; formal training is not required.
D	**Importance.** Task is not important; it has no significant impact on mission success.
E	**Frequency.** Task is rarely performed, no formal training is required.
F	**Lack of equipment.** Currently there is no equipment in the CAF inventory to train this task.

KEY:

TASK	KNOWLEDGE	SKILL	ATTITUDE

Task No	Statement	T/NT	PO Number	Rationale
XT0001	Configure KY-99M to establish communication	T	nil	nil
XT0002	Load keymat into KY-99M equipment IAW KY-99M SOP	T	001	nil
XT0004	Complete over the air distribution (OTAD) of KY-99M	NT	nil	Performed infrequently on the job, and SOPs are available to assist in this performance
XT0005	Determine circuit requirements for operation from the OpTask Com	T	001	nil

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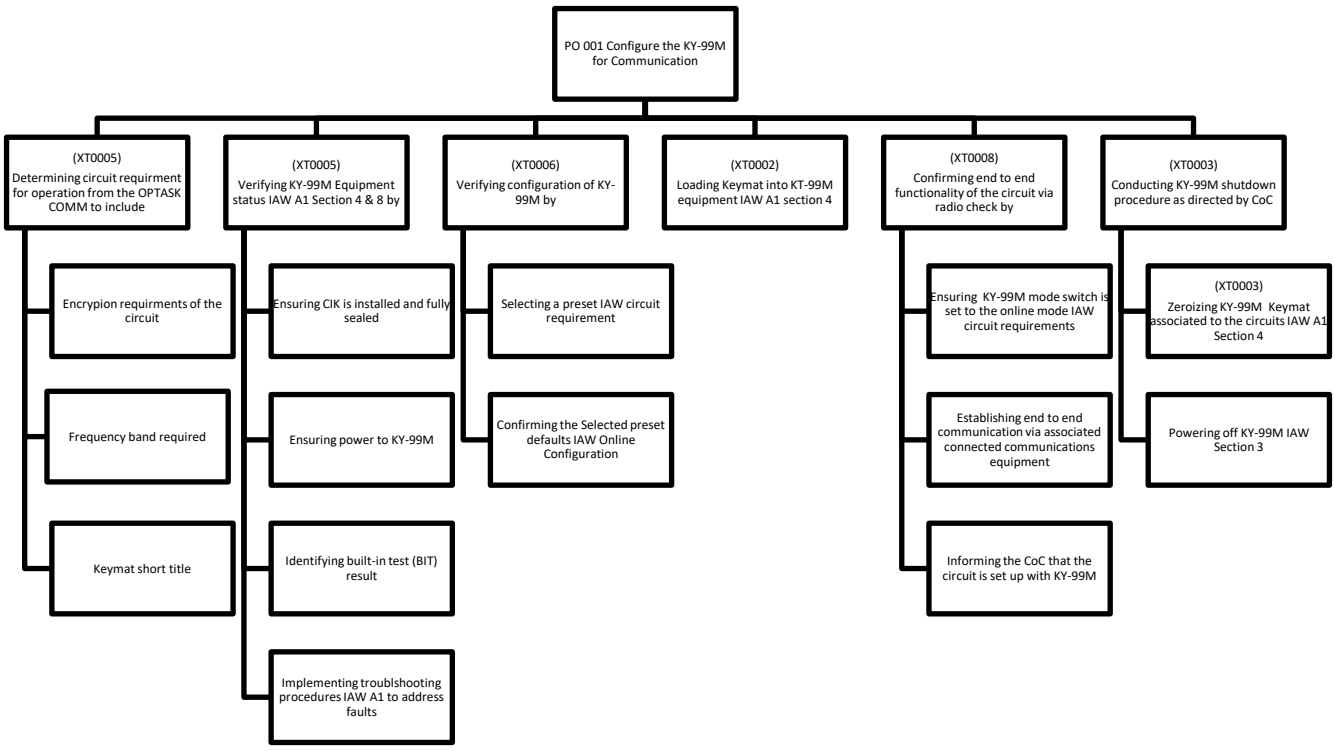
Task No	Statement	T/NT	PO Number	Rationale
XT0006	Verify configuration of KY-99M	T	001	nil
XT0007	Verify KY-99M equipment status	T	001	nil
XT0008	Confirm end-to-end functionality of the circuit via radio check	T	001	nil
XT0009	Conduct KY-99M shutdown procedure	T	001	nil
XK0001	Identify Operational and Configuration Management Interface (OCMI) menus	T	nil	nil
XK0002	Keymat fill device applications	T	nil	nil
XK0003	Explain fundamentals of military frequency bands	T	nil	nil
XK0004	Differentiate between plain and secure communications	T	nil	nil
XK0005	Explain cryptographic fundamentals	T	nil	nil
XK0006	Identify KY-99M equipment	T	nil	nil
XK0007	Explain the purpose of the KY-99M	T	nil	nil
XK0008	Explain power requirements of the KY-99M	T	nil	nil
XK0009	Explain built-in test (BIT) for the KY-99M	T	nil	nil
XK0010	Explain loopback test for the KY-99M	T	nil	nil
XK0011	Explain over the air distribution (OTAD) specific to the KY-99M	T	nil	nil
XS0001	Input information into graphic user interface (GUI)	T	nil	nil

Task No	Statement	T/NT	PO Number	Rationale
XS0002	Input information to liquid crystal display (LCD)	T	nil	nil
XS0003	Utilize keymat fill devices	T	nil	nil
XS0004	Navigate the KY-99M menu tree (GUI)	T	nil	nil
XS0005	Navigate the KY-99M menu tree (LCD)	T	nil	nil
XS0006	Change battery in the KY-99M	T	nil	nil
XS0007	Power on the KY-99M	T	nil	nil
XA0001	Security awareness (INFOSEC/COMSEC)	T	nil	nil

ANNEX C – QS SCALAR

PO 001



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ANNEX D – RECORD OF DECISIONS QSWB

1. This record summarizes the conduct of the QSWB including major activities, important discussions and contentious issues, as well as all decisions and recommendations made by the Board. A description of the reasoning which supports a decision or recommendation is provided when required.

Title	**KY-99M Operator QSWB**			
Location of Board	**CFB Esquimalt N50**			
Dates	**8-11 Aug 2023**			
Board Composition				
	Surname	Initials	Rank	Unit
Co-Chair (content)	Pinard	S	Lt(N)	NFS(P)
Co-Chair (process)				
TM	Mason	C	Capt	NTG HQ
SME	White	D	PO1	NFS(P)
SME	Ivey	R	PO1	CANFLTPAC HQ
SME	Gill	J	MS	HMCS PROTECTEUR
SME	Quesnelle	L.M	PO2	NPTG NTDC (A)
Secretary	Gauthier	F	A/Slt	HMCS VENTURE
Advisor	Schultz	M.J	CPO2	NTG HQ

Serial	Discussion	Action
	Day 1: 08 Aug 2023	
1	The board chairperson opened by providing a short brief of their background and expectations for this specific QSWB. They also outlined the names and experience of each person participating in the board, and their previous experience relating to writing boards. Roles, guidelines, and responsibilities of the board members were also discussed.	
2	Following the introductions, an outline of the purpose of the board was highlighted by the board chair. The aim of the board is to produce a qualification standard (QS) that is reflective of the job (Configuring the KY-99 crypto end unit).	
3	TDO brief outlining the QS board process for the KY-99 crypto end device, the use of analysis in the board, and an overview of	

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	Canadian Forces Individual Training and Education System (CFITES) as it relates to the process for this board.	
4	TDO asserted that the board was strictly at the analysis stage of the CFITES cycle, and the board was to define the job in terms of operation performance without delving into the training side of the job. Board needs to develop task list relating to the KY-99 end crypto unit and its role in the job and finalize QS.	
5	Discussion commenced with all on-site participants on the specificity required when defining a task list for the job and defining the job itself when developing the QS. History of the end crypto unit and associated ICTs (Initial Cadre Training) that occurred (3 conducted – Ottawa, Halifax, Esquimalt) and 2 more planned for autumn 2023 (Halifax and Esquimalt). There was discussion over whether the job would ever include maintenance of the end crypto unit (at the unit level) or just its operation, which led to further task list considerations. A SME (Subject Matter Experts) pointed out that, due the nature of the end crypto unit and its design, should a KY-99 fail, another one in the rack would be repurposed for that circuit and the failed KY-99 would be returned to 76 Comm Group for repairs.	
6	Points discussed between on-site participants were reiterated to pers attending remotely. The need for the RCN (Royal Canadian Navy) to control their own QSPs for specific end crypto units currently controlled by the Canadian Army, was noted (TACLANE and KIV-7M), and whether the currently utilized Canadian Army QSPs could be replaced or improved upon for the RCN's purpose. RCN becoming the TA (Training Authority) for these items. The need for safe-to-fail training for NAV COMMs was noted and compared with other trades that have similar training. The need for practical hands-on training was identified as a "must" to increase proficiency and confidence. Highlighted need for HQ to	

	support training and understand historical QS board processes to streamline decision making.	
7	Need for concrete definitions of POs (Performance Objectives), TPs (Training Plan), and QSPs was discussed, as well as the desire for more pers to understand the decision-making process so that pers understand why certain objectives and training points were agreed upon during the conduct of QS and/or TPWB.	
8	Document “KY-99M Modern Secure Radio Voice and Data Encryptor version 9.0” was introduced by the Training Manager (TM). This document covers the use of the KY-99 system and its maintenance and was created by the contractor delivering the ICT.	
9	SMEs raised questions on what type of course is being developed (operator only, maintainer only or operator/maintainer). Decision was made that this board was focused on the operator side only, as the maintenance of the end crypto unit will be handled by FMF or through 76 Comm Group.	
10	Chairperson raised question regarding the current availability of comprehensive Standard Operating Procedures (SOP) that describe various failed state of the equipment and frontline resolutions.	
11	The TDO guided the participants through Rapid Targeted Population Analysis for who will be intended to attend KY-99 courses delivered by the RCN - Population was identified as S2/S3 (pers completely new to trade). Expected: -certain aptitude/interest for IT knowledge - associated CFAT scores - basic, if any, knowledge of crypto (likely none) - no understanding of how individual pieces of kit fit into the overall picture - requirement of fundamental CIS knowledge - No training in equipment for QL3 and below - ages 17-57	

12	SME noted concern over giving this course so early in the trade when pers are not expected to have a fundamental knowledge of trade specifics (e.g., frequency bands). Chairperson asserted that the board's intent is to solely tackle how knowledge of this equipment will be taught, not where that training exists within the trade progression. SME noted that use of this equipment at the S2/3 level would be under supervision, and this equipment would eventually be a staple of the trade. -SME noted concerns that pers could operate equipment without understanding the bigger picture of what the kit does in the trade. Noted it behooves course instructors to explain the kit and its place within the within the CISN infrastructure so that a wider understanding is held by members of the trade. TDO noted importance of potentially modifying current ICT courseware to include information on why this kit is being utilized over other kit, and its larger purpose by relying on relating to expected previous knowledge of members (adult learning considerations). - Chairperson noted that a recommendation can be placed within the assessment of this piece of kit (performance objective) that this course be placed in the training sequence after fundamentals have been taught to better integrate the knowledge of the kit into the larger trade training.	
13	TDO presented slide show on Task Disposition- to better capture whether each task presented by SME (10 tasks) would be 'train or no-train'. To create criteria for trainable tasks Provide rationale if no training to be provided 3 methods; gut feeling, DIF method, and criterion approach. Things in support of no-train designation are previous qualifications, expected future training, low frequency of use, low importance, or a lack of equipment for training (no longer used in CAF (Canadian Armed Forces)).	
14	SME pointed out a potential requirement for CISOs to be delivered this training in the conduct of their COMSEC custodian duties. The TM indicated that the current CISO QSP infers that	

	they are trained on how to use end crypto units (PO 006). SME noted that CISO training is more geared to KG 136. Chairperson stated that CISOs do not require knowledge of the use of the KY-99, only operating as an intermediary issuing the end crypto unit. They also indicated that frequency at which CISO would be expected to operate the KY-99M is so low as to make the training redundant in a CISO course. Noted that CISO training is more about COMSEC. The chairperson indicated that Snr Nav Comms should be participants in QS/TP Review board for the CISO courses, to ensure that POs accurately reflect what a CISO should do, so as not to overlap with what Nav Comms are expected to perform. TDO indicated that CISOs may need to know what the kit does, but not how to operate it. Also noted that training would only be useful if it is used on the job, and that providing operation training to CISO and similar would be a waste of time and resources. Board agreed that specific knowledge of operating the kit is only necessary to those who are operating it, and as such only those pers will be considered within the board (NAVCOMMs, excluding CISO and engineering Public Service employees).	
15	SME raised question over whether KY-99M would be solely for naval use, or if training should be made available to other elements. Board decided that inclusion of multiple trades and elements in the training should be considered for the MESCR. The Chairperson recommended that this be clearly articulated in the prerequisite portion of the QS.	
16	TDO presented document of 10 tasks relating to the operation of the KY-99. Board initially reviewed tasks from the NavComm MESIP, it was later decided to create a new task list as this was deemed common training and therefore should have a distinct task list for the MESCR to DPGR to create an NQual for CSQ. TM indicated that he gathered all possible tasks that he deemed met the intent for the operation of the KY-99.	

	AT0060 setup communications circuit onboard ship - Deemed by board to be no-train, this is a core part of previously delivered training. AT0090 Set up HF comms circuit onboard ship DT0215 Deemed by board to be NT as it falls under comms circuit training, previously trained. DT0215 Load keymat into crypto kit onboard ship Train DT0235 update crypto equipment with transfer devices NT, falls under SKL (single key loader) training DT0240 update crypto daily on ship NT, is covered under DT0235 DT0300 Destroy crypto Keymat on ship - Train- SME/TM indicated training is required to reinforce previous training and is critical based on COMSEC policies/regulation, and zeroing end crypto unit as a form of destruction of the keymat.	
17	DT0315 configuring crypto equipment Train. Chairperson indicated the necessity of clearly articulating what “configuring crypto equipment” means. SMEs indicated that configuring the end crypto unit would have to do with considering configuration sheets for the KY-99 or GUI specific to device, or LCD tree menus. Board ultimately decided to train, as this information is specific to the KY-99M device. Decided to train with LCD (on the device), but GUI (3 presets in the KY-99M) after LCD training would be most intuitive. Determined that both LCD and GUI should be trained.	
18	DT0320 complete over the air rekeying (OTAR) of crypto material onboard ship Board deferred to SMEs for decision.	

	SMEs argued to train this task, as it is vital to the operation of the end crypto unit, however SME also noted importance of where OTAR is coming from, complicating how OTAR would be taught in conjunction with OTAT and OTAD. Training would then require another ship that is available and knowledgeable in the training to conduct training. Board determined that OTAR is conducted rarely and could be satisfied by a job aid, but that it should be trained as OTAR, as this process is significantly different between the various end crypto units.	
19	DT0395 restore crypto equipment after power failure on ship Board determined NT, as this would be covered under LCD/GUI training (DT0315). SMEs indicated that new equipment on ships makes power failure duration, short and occurrences, rare.	
20	S00160 set up voice circuit No Train. Covered under IWS PO 002	
21	SMEs raised question on whether a task should be added for emergency destruction of keymat/end crypto unit. SME noted CAF does not train to emergency destruction, as that process is up to based on SOPs. SME discussed destruction as a physical process of damaging kit in a specific way, rather than just a process which includes the destruction as part of a larger process. Board decided training would be useful in determining how this would apply to the KY-99 specifically, and to craft this training based upon established destruction policies for COMSEC material/end crypto unit.	
	Day 2: Aug 09 2023	

22	LCdr Peever got back to TDO regarding process for creating the CSQ with DPGR and confirmed what had been previously relayed	
23	Chairperson indicated that it would be beneficial to keep the training generic, so that the training can be adapted to other trades/end crypto units without having to wait for new specific training to be developed or QS/TP approved.	
24	TDO presented slideshow brief on task analysis They emphasized task list being solely related to on-the-job requirements, specifying that tasks must be observable and measurable	
25	TM presented list of tasks from the MESIP that were deemed by board to need training, tasks were rearranged, and sub-tasks added as deemed necessary	
26	SMEs suggested putting everything under 1 PO- Operating the KY-99M, or in 2 POs- Operating the KY-99M and OTAR. Chairperson asked if one could configure the device without using it- then suggested that 2 POs be configuration (DT0315) and operation of KY-99M. Board discussed place of OTAR procedure for the KY-99M fits into the suggested POs	
27	TDO suggested zeroizing KY-99M be part of the configuration (DT0315) PO, Board concurred. Suggested steps of configuration: Determine circuit (UHF, HF) IAW OpTask Comm Choose associated preset through LCD or GUI IAW circuit requirements Verify preset settings IAW config sheet DT0215 load keymap (TEK, KEK, periodic loading d/w/m/y)	

	Considered that under supervision, it can be understood what is being asked OpTask as an input - can be presented as a teaching tool/point - possible lesson activity to extract information (but may overlap with S3/S2 Course – so may serve as a review) SME noted you do not need OpTask at this level, but it would be beneficial to familiarize with OpTask Board members recommend cfcd107 rewrite to capture all parameters that all crypto units should be set at IOT effectively use them in all required configurations Discussed whether common operation will be done through GUI or LCD- determined most common will be GUI, need LCD in case GUI unavailable SME suggested teaching configuration to GUI, adding LCD post-training on GUI	
	Day 3: Aug 10th, 2023,	
28	Board noted that current Crypto trainer may require update as there are no radios in the current trainer. Current trainer has capability to network with radios, it is not integrated in an overall comms suite, however, there is a way to prove the circuit end-to-end functionality in the trainer. Therefore, while the training limitation is that the actual OTAD cannot be performed in house, the proof of the circuit enables trainees to complete the procedure outlined and will sufficiently prepare them to do it in real world scenarios. SME spoke to the fault-finding processes are in the SOP and error codes on the KY-99M are displayed and searching in the SOP is intuitive. TDO suggested radio check as one of the final steps in the PO as a means to prove the circuit Board discussed potential need of additional kit (radios) from manufacturer (Raytheon) IOT conduct training a “nice to have”	

	as it would provide more realistic training environment - however each element uses different kit. Board discussed that OTAR does not need to be its own PO, based on knowledge in the trade of OTAR on other devices Suggested it could be an EO/EC/PC as part of a comms check PO Suggested OTAR and comm check could be used as confirmation within the Configuration PO	
29	Board decided OTAR is NT- removed from list of POs – determined only 1 performance objective will be to configure the circuit and prove it works via radio check .	
30	Continued discussion on configuration PO. Amended UHF/HF and plain/secured determinations as part of determining CCT EO- noted these points should remain as teaching points, but fall under determination of the circuit as a whole Discussed config sheets, Board chair recommended these config sheets be part of CFCD107- SMEs concurred SME noted CFCD might only contain config sheets for things BIS owns Recommend - KY-99M config sheets be added to CFCD107 and suggested review of full revision of CFCD107 (DNIW is the authority / contact).	
31	SMEs revisited building a circuit- added verify cryptographic ignition key (CIK) SMEs identified that trainees do not require comprehensive knowledge of TEK or KEK IOT load the keymat Board decided on DT0320 (from Occ Spec) OTAR- happens infrequently therefore also does not require training, also noted this would not be the job of an S3 or equivalent	

	Discussed making zeroizing the device a final step in the configuration PO to ensure there is no residual crypto on the device -SME noted this would be displayed by the device upon powerup, based on this information, SME discussed that best practice would be to zeroize as a first step if the device displays preloaded crypto SME noted that zeroizing device could create errors with the device Board identified concerns with too broad a description of zeroizing procedures during training may result in inadvertent COMSEC incidents – trainees need to be fully aware of what they are zeroizing and the implications on COMSEC and operations if this is done incorrectly. ADM(IM) CIS Publications - KY-99A Operating Instructions (CIS 01417) – may be a good reference to assist with building this course. (Note to TP Board) Board determined that zeroizing be removed from the suggested steps, replaced with confirming there is no key when the device powers on, added as part of power on check (light comes on, KY-99M LCD displays information, does BIT) Decided to park DT0300 – Destroy Cryptographic Keymat Replaced with zeroize specific keymat only	
32	TM presented PowerPoint on the powering up of the KY-99M	
33	Discussed being able to modify language in the Raytheon user guide to work within the framework of the CAF as a training / job aid SME suggested steps could be confirming BIT test with no errors, troubleshoot if there are errors, troubleshooting may result in switching to another KY-99M	

	Discussion on what levels on information should be contained within the POs- if all will be taught under supervision, self-taught or classroom, etc. Chair of board noted device is from 1994, has been extensively used by the US- common problems re well-documented	
34	Based on DPGR confirmation that the task list will be created by the board and submitted via MESCR – board utilized the QSMS system to begin creation of task list including tasks, skills, knowledges and attitudes. Board remained neutral on element and created the task list with CSQ framework in mind. Discussed adding a teaching point about why destruction is not taught as involves physically damaging equipment XT0004: Complete OTAD of KY-99M (NT- rationale infrequent use and SOPs are available to perform task) Knowledge, Skill and Attitude element added:	
35	Defined security level of the whole circuit whether it is plain / secured	
36	TDO noted need for the TP board to (teaching point / lesson) reinforce the importance of battery replacement in KY-99M. Complacency may lead to deactivation of KY-99M, rendering it inoperable. This was a reported issue from ICT course. SOP identifies the 1-minute limit for swapping it out and 5-year cycle as a best practice for battery replacement.	
37	TDO noted that for PO001.4 all the processes were in the SOP and as such were not noted in the PO.	
	Day 4: Aug 11, 2023	
38	Board tidied up and recorded PO information discussed previously, organized appropriately, added prerequisites, clearance needs to take training	
39	CCD86 identifies that the KY-99M is unclassified, therefore the prerequisites accurately reflect the requirements of this course	

40	During discussion of prerequisites the TDO recommended that COMSEC and INFOSEC courses should be a consideration. The board determined that personnel who are posted to those environments would be required during onboarding to attend COMSEC / INFOSEC training	
41	Cryptosecurity Standard (mil.ca) For references. DND/CAF Communications Security Standard (CSS) for Controlled Cryptographic Items Managed by the Defence Supply Chain – CSS-20	
42	KY-99M SOP Version 9.0.docx (mil.ca) For references: KY-99M Modern Secure Radio Voice and Data Encryptor Version 9.0	
43	OpTask Comm is a NATO standard, and will be used in this PO	
44	Board Chair expressed concerns about the Personnel Security Clearance requirement written as a prerequisite for selection for this training. However, after further investigation and reaching out to NFS(P) USS, the following is clarification of the issue and resolution: Crypto training rooms are designated as Operations Areas, which restricts access to specific pers based on Access Lists approved by the TEs' Commandants. Students can access these trainers, if they are supervised by pers that have been granted access through these Access Lists; and The classification of these trainers is fluid, based on the classification of the keymat filled/loaded in end crypto units. As long as the keymat remains unclassified, the minimum-Security Clearance to access the space is Enhanced Reliability (ER).	
45	Supervision is implied as part of the CAF based on rank level-junior pers are supervised by more senior pers	
46	SOPs can change across trades/units, may change operation of KY-99M unit.	
47	Conducted AAR	

RECORD OF AMENDMENT & AMPLIFICATION

1. The record of amendment and amplification will be filled in whenever a change to the QS is initiated. This section is used to provide amplification details and provide a history of the changes made. In the requirement for the amendment section, the member inputting the changes will transpose the information from the submitted QS Amendment Form, thus ensuring the QSP has the most relevant information.

Version:	**X.X**
Date:	**Who Initiated Change:**
Requirement for the Amendment:	
Description of Changes:	
Change Tracking number:	
Trg Mgr:	

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TRAINING PLAN ANNEXES

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ANNEX E – RESOURCES, EQUIPMENT, SUPPLIES, AND ANCILLARIES

1. The following resources, equipment, supplies, and other elements are required in support of training. Resources are described in terms of formed organizations and are to arrive equipped with all personnel, vehicles, weapons, communications, and ancillary equipment required for conduct and integral support of operations described in the POs.

2. Facilities/Trainers

SER	DESCRIPTION		QTY	REMARKS
a.	Classified Classroom	Cleared for COMSEC equipment	1	
NOTES:				

3. Ammunition: Nil

4. Weapons: Nil

5. Specialized/Technical Equipment: Nil.

6. Transport/Vehicles: Nil.

7. Communication Equipment/Frequencies

SER	NSN	DESCRIPTION		QTY	REMARKS
a.	5810-01-617-4671	KY-99M		12	
b.	6130-01-548-1505	SKL		12	Loaded with training crypto
c.	Any	Laptop		12	
d.	5810-01-066-7587	Fill cable		12	
e.	Any	Cat 5 cable		12	
f.	Supplier	MKJ1A adaptor		12	
g.	Any	handset		12	
NOTES:					

8. Training Aids: Nil

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9. POL Products: Nil

10. General Purpose Stores: Nil

11. Stationery

SER	DESCRIPTION		QTY	REMARKS
a.	Pencils		24	
b.	Pens		12	
NOTES:				

12. Personal Equipment: Nil

All students will have in their possession the personal equipment detailed in the course joining instructions. All personal equipment to be assigned to the students are accounted for in the resource tables above.

ANNEX F – COSTING INFORMATION

Estimates only, based on CF Cost Factors Manual

cost-factors-manual (mil.ca)

****5 year total based on one course per year INCLUDING Initial Set-Up. Costs may be adjusted accordingly.**

Course: KY-99M Operator (20 courses per year)		Serial:				
TD	No. Of Students / instructors /	Cost per each	# Days (if applicable)	Initial TE Set-up	Each successive course	5 Year total
Airfare (Return)	2/12	$1,000.0		$2,000	$2,000	$ 200,000
R&Q	2/12	$75.0	5	$750	$750	$ 75,000
Incidentals	2/12	$17.5	5	$175	$175	$ 17,500
Subtotal				**$2,925**	**$2,925**	**$ 292,500**
Material	Quantity	Cost per each	# Days (enter 1 if not applicable)			
PP&S (Estimate)	12	$50.0	1	$600	$600	$60,000
KY-99M	12	$14,427	1	$173,124		$173,124
SKL	12	$1,600	1	$19,200	$0	$19,200
Fill cable	12	$120	1	$1,440	$0	$1,440
Cat 5 cable	12	$30	1	$360	$0	$360
MKJ1A adaptor	12	$600	1	$7,200	$0	$7,200
handset	12	$400	1	$4,800		$4,800
Instructional Materials (Printed / Handouts etc.)	12	$30.0	1	$450	$450	$45,000
Subtotal				**$207,174**	**$1,050**	**$311,124**
PYs (only if course includes incremental on SWE, or course to be delivered off site	Quantity	Cost per day* (SWE if necessary)	# Days (if applicable)			
N/A	1	$	1	$0	$0	$0
		subtotal		**$0**	**$0**	**$ 0**
	Total Estimated Costs			**$210,099**	**$3,975**	**$603,624**

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ANNEX G – SUGGESTED SEQUENCING AND TIMINGS

1. This annex is to be used to properly schedule all required classrooms, trainers, ships, personnel, etc. for the duration of the course. Admin and PT periods are accounted for IAW NPTORD 4500-22 Annex B (QSP Instructions).

2. The tables are completed in the preferred sequence for lessons and resources. Special instructions to schedulers shall be included as "Notes" following the applicable PO or Module.

3. The table below is available to use as a template for calculations.

Allocation of ATT	**Minutes**
Actual Training Time	1000
ATT Sub-total	**1000**
Allocation of BAT, TAT, TTT, PT (Course > 31 days)	
Base Administrative Time (BAT)	50
Training Administrative Time (TAT)	50
Training Travel Time (TTT)	0
Physical Fitness (PT)	0
Sub-total	**100**
Course Duration Calculation (ATT+BAT+TTT+TAT+PT) **Total mins/400 = # days** (rounded up)	
Total Training Days	**3 days**

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OPTIMUM TRAINING SCHEDULE

Week 1 Time	Day 1	Day 2	Day 3
Period 1 0800-0850	**EO 001.01** Initialize KY-99M for operational use	**EO 001.02** Load keymat into KY-99M equipment	**TAT**
Period 2 0850-0940	**EO 001.01** Initialize KY-99M for operational use	**EO 001.02** Load keymat into KY-99M equipment	**EC 001**
Break			
Period 3 0950-1040	**EO 001.01** Initialize KY-99M for operational use	**EO 001.02** Load keymat into KY-99M equipment	**EC 001**
Period 4 1040-1130	**EO 001.01** Initialize KY-99M for operational use	**EO 001.02** Load keymat into KY-99M equipment	**EC 001**
Lunch			
Period 5 1230-1310	**EO 001.01** Initialize KY-99M for operational use	**EO 001.02** Load keymat into KY-99M equipment	**PC 001**
Period 6 1310-1410	**BAT**	**EO 001.02** Load keymat into KY-99M equipment	**PC 001**
Break			
Period 7 1420-1510	**EO 001.02** Load keymat into KY-99M equipment	**EO 001.02** Load keymat into KY-99M equipment	**Re-test PC 001**
Period 8 1510-1600	**EO 001.02** Load keymat into KY-99M equipment	**EO 001.02** Load keymat into KY-99M equipment	Re-test PC 001

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ANNEX H – ASSESSMENT PLAN

EC / PC	Scope	Purpose	How	When	Time	Resources	Limitations	Pass/Fail Criteria	Impact
EC 001	EO 001.01 And EO 001.02	Initialize KY-99M for operational use And Load keymat into KY-99M equipment	Trainees will configure the KY-99M for use and answer written troubleshooting questions.	Upon completion of EO 001.02	150 min	Classified Classroom	Troubleshooting scenarios cannot be replicated.	Practical run: 70% Written assessment: 80%	Remedial training.
PC 001	PO 001	Configure the KY-99M for communication	Trainees will configure the KY-99M for use	Upon completion of EC 001	100 min	Classified classroom	Troubleshooting scenarios cannot be replicated.	90%	1st failure - Re-test IAW NPTORD 4500-54 2nd Failure - TRB IAW 4500-54

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ANNEX I – ASSESSMENT CHECKLISTS

ENABLING CHECK 001.01

EC **Time: 120** mins (concurrent) + 30 min debrief

ASSESSMENT ACTIVITY:

In the classified classroom, the trainees will be provided a KY-99M device, a scenario with operational requirements and the KY-99M SOP.

The trainees will be required to successfully configure the KY-99M for operational use. Operational use is considered to be reached when the device can transmit and receive information with another device on a narrowband channel and a wideband channel. Simultaneously they will also be required to answer 10 written questions on trouble shooting and general knowledge.

At specific confirmation checks, the trainees will have to raise their hands for the instructor to come verify before they move on to the next step.

The instructor will supervise all trainees at the same time and assist them when needed by prompting them in the right direction. The instructor will gather all written tests for correction at the end of the EC.

Feedback will be conducted as a group to cover common mistakes that occurred throughout all trainees' run. Individual feedback should be used as required.

Pass/fail criteria:
Practical run: 19/26, 70%
written part: 8/10, 80%

ASSESSMENT RESOURCES:

- KY-99M
- KY-99M SOP
- Ancillary equipment
- SKL with training keymat
- Scenario
- Laptop
- Written exam
- Pen and pencils

IMPACT OF FAILURE: Remedial Training.

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INVIGILATOR NOTES:

(KY-99M Operator) – EC 001.01 Assessment Checklist			
Trainee Name:	**Assessor Name:**	**Date:**	**Session:**

1. PASS / FAIL CRITERIA: 19/26

2. IMPACT OF FAILURE: Remedial training.

ITEM	PASS	FAIL	COMMENTS
1. Verify KY-99M equipment status			
Ensure CIK is installed and fully seated: - CIK serial number matches with KY-99M - Security pin has locked the CIK in place			
Ensure power to KY-99M by turning selector/mode switch to “PT”;			
Confirmation check: The LCD display “PT … No keys”			
2. Verify configuration			
Confirmation check: Change color of screen to red through offline menu			
Prepare OCMI to configure ECU settings			
Enable OCMI within the ECUI			
Turn Selector/mode switch to “remote”			
Connect computer to the KY-99M using the OCMI port			
Modify NIC set up to allow connection through OCMI port IAW A1 Sect 11.1			
Confirmation check: Logging in to maintainer account through Web Browser			
Configure settings in OCMI IAW A1 sect 11 and operational requirements from OpTask Comm			
3. Load keymat			
Transfer keymat from SKL to KY-99M in a profile load			
Turn Selector/mode switch to “offline”			
Connect the SKL			

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ITEM	PASS	FAIL	COMMENTS
Turn on SKL and log in			
Follow A1 sect 4.7 steps to load narrowband channel			
Follow A1 sect 4.7 steps to load wideband channel			
Disconnect fill cable from KY-99M			
Confirmation check:			
Report successful end-to-end functionality of the narrowband circuit			
Report successful end-to-end functionality of the wideband circuit			
4. Zeroize KY-99M keymat			
Ensure Selector/mode switch to “OFL”			
Zeroize a single key IAW A1 4.6			
Confirmation check: Confirm zeroization procedures by entering Key Ops menu to view any remaining keys.			
Zeroize all keys			
Turn selector/mode switch all the way clockwise pulling out to “Z all”			
Monitor ECU for power cycle instructions			
Cycle power			
Confirmation check: Confirm zeroization procedures by entering Key Ops menu to view any remaining keys.			
5. Shut down KY-99M			
Power off KY-99M			

ADDITIONAL COMMENTS	
STRENGTHS:	**AREAS FOR DEVELOPMENT:**

THE TRAINEE HAS RECEIVED A DEBRIEF AND SEEN THE CONTENT OF THIS ASSESSMENT			
OVERALL ASSESSMENT (circle one)		**FAIL**	**PASS**
Assessor	Name:	Date:	Signature:
Trainee	Name:	Date:	Signature:

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PERFORMANCE CHECK 001

PC **Time: 80** mins (concurrent) + 20 min debrief

ASSESSMENT ACTIVITY:

In the classified classroom, the trainees will be provided a KY-99M device, a scenario with operational requirements and the KY-99M SOP.

The trainees will be required to successfully configure the KY-99M for operational use. Operational use is considered to be reached when the device can transmit and receive information with another device on a narrowband channel and a wideband channel.

The instructor will supervise all trainees at the same time and will conduct the radio check with each student.

Feedback will be conducted individual feedback.

Pass/fail criteria: 7/8, 90%

ASSESSMENT RESOURCES:

- KY-99M
- KY-99M SOP
- Ancillary equipment
- SKL with training keymat
- Scenario
- Laptop
- Written exam
- Pen and pencils

IMPACT OF FAILURE: Re-test IAW NPTORD 4500-54

INVIGILATOR NOTES:

(KY-99M Operator) – PC 001 Assessment Checklist			
Trainee Name:	**Assessor Name:**	**Date:**	**Session:**

3. PASS / FAIL CRITERIA: 7/8

4. IMPACT OF FAILURE: Re-test IAW NPTORD 4500-54

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ITEM	PASS	FAIL	COMMENTS
1. Verify KY-99M equipment status			
Ensure CIK is installed and fully seated: - CIK serial number matches with KY-99M - Security pin has locked the CIK in place - Ensure power to KY-99M by turning selector/mode switch to “PT”;			
2. Verify configuration			
Prepare OCMI to configure ECU settings: - Enable OCMI within the ECUI - Turn Selector/mode switch to “remote” -Connect computer to the KY-99M using the OCMI port -Modify NIC set up to allow connection through OCMI port IAW A1 Sect 11.1 -Logging in to maintainer account through Web Browser -Configure settings in OCMI IAW A1 sect 11 and operational requirements from OpTask Comm			
3. Load keymat			
Transfer keymat from SKL to KY-99M: -Turn Selector/mode switch to “offline” -Connect the SKL -Turn on SKL and log in -Follow A1 sect 4.7 steps to load narrowband channel and wideband channel - Disconnect fill cable from KY-99M			
Confirmation check:			
Report successful end-to-end functionality of the narrowband circuit			
Report successful end-to-end functionality of the wideband circuit			
4. Zeroize KY-99M keymat			
Zeroize the device			
Confirmation check: Confirm zeroization procedures by entering Key Ops menu to view any remaining keys.			
5. Shut down KY-99M			
Power off KY-99M			
ADDITIONAL COMMENTS			

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ITEM	PASS	FAIL	COMMENTS

STRENGTHS:	AREAS FOR DEVELOPMENT:

THE TRAINEE HAS RECEIVED A DEBRIEF AND SEEN THE CONTENT OF THIS ASSESSMENT			
OVERALL ASSESSMENT (circle one)		FAIL	PASS
Assessor	Name:	Date:	Signature:
Trainee	Name:	Date:	Signature:

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ANNEX J – ON JOB PERFORMANCE RECORD (OJPR)

Not applicable to this course.

Version	Revision Date	Remarks
OJPR Version 1.0	DD / MM/ YYYY	New OJPR

Name and Initials	Rank	Service Number

Date Training Started	DD MMM YYYY
Date of Completion	DD MMM YYYY

<u>THE OJPR MUST BE COMPLETED IN ACCORDANCE WITH THE INSTRUCTIONS BELOW. INCORRECT OR INCOMPLETE OJPR WILL BE RETURNED FOR RE-SUBMISSION.</u>

INSTRUCTIONS

1. One copy of this OJPR, hereinafter referred to as the master copy, shall be made up for each individual beginning OJT. It shall be kept and maintained by the supervisor. The Training Establishment (TE) shall make a duplicate copy for retention by the trainee for the purpose of recording and reviewing individual progress. The TE for the OJPR must maintain Training Tracker fields as outlined in Monitor Mass.

2. As the trainee completes each Performance or Enabling Check to the required standard, IAW the associated checklist in ANNEX I, the Training PO or Officer will date and sign the applicable item on the master and duplicate copies of the OJPR sheet and inform the individual of the action taken.

3. It may not be possible to conduct OJT for some POs if, for example, the necessary equipment is not fitted on a ship, is not on issue to the unit, or is otherwise unavailable. If the training PO or Officer encounters any of these issues they are directed to contact Standards at the appropriate TE for guidance.

4. If an individual demonstrates the ability to perform additional duties not specified in this QSP, an appropriate entry shall be made under "Additional Duties" at the end of the OJPR.

PROGRESS REVIEW

5. Progress review shall be carried out quarterly by the Divisional Officer and Section Head (NCMs) or Training Officer (Officers).

Example OJPR record:

PO/EO	Description	Remarks	Rank/Name/ Signature	Date	OPI and Unit
PC/EC XXX.XX				DD/MM/YYYY	

OJPR Reviewed By OJT Supervisor (Rank, Position, Name)	Signature	Date
Section Head (NCMs) or HOD (Officers)		

TRAINING COMPLETION CERTIFICATE

Date training completed:	
Div Officer (NCMs) or CO (Officers)	Date

CERTIFICATION OF QUALIFICATION

I certify that _____ has completed the OJT to the standards stated in the Name of QSP.

TE Cmdt/CO	Date

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ANNEX K – TT PLAN

EO	COA 1 TEL Delivery	COA 2 Interim Delivery	**Impact if TEL not implemented** - Realized benefits - E/W Coast Delta
Nil.			

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ANNEX L – RECORD OF DECISIONS TPWB

1. This record summarizes the conduct of the TPWB including major activities, important discussions and contentious issues, as well as all decisions and recommendations made by the Board. A description of the reasoning which supports a decision or recommendation is provided when required.

Title	**KY-99M Operator TPWB**			
Location of Board	**NFS(A), CFB Halifax**			
Dates	**03-13 Oct 2023**			
Board Composition				
	Surname	Initials	Rank	Unit
Co-Chair (content)	Kourzenkov	I.	Lt(N)	NFS(A)
Co-Chair (process)	Martin-Labelle	J.C.M.	SLt	NFS(A)
SME	Laplante	D.F.	MS	NFS(A)
SME	Stone	J.R.	MS	NFS(A)
SME	Olizarevitch	M.L.	S1	HMCS St. John's
SME	Bunleon	K.K.	PO2	NFS(A)

Item	Details	Action
1	Target Population Analysis: - This course will be taken by members who are OFP qualified from an operational unit or a shore unit, and maybe by some maintainers. - This course is applicable to the following occupations: ATIS TECH, NAV COMM, CYBER OP, SIG OP, SIG TECH, and IS TECH. - Most members have limited time to come on this course due to their primary job at their units. - Ideally, the number of students match the number of equipment available (8 on the East Coast and 8 on the West Coast) - They have basic to intermediate level of knowledge of the SKL and GUIs. - Students learn best with hands on application, ideally we balance theory and practice throughout as opposed to delivering all theory then practicing on the piece of equipment.	
2.	"Selector/mode switch" refers to the switch on the left of the console and the term "VOL/DSP/PNL knob" refers to the smaller switch on the right of the console.	

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Item	Details	Action
3.	The board debated whether or not to have an entire EO dedicated to troubleshooting because there are a lot of knowledge required to recognize where the error occurred during the entire process of configuring the KY-99M for use. In the end, the board decided to inject standards of troubleshooting procedures throughout the configuration steps. Reasoning is that, during the course students will most likely experience errors as they practice configuring the device and it would be more valuable to go over errors at each step of the way while they happen. The board decided on two EOs: EO 001 – Initialize KY-99M for operational use EO 002 – Loading keymat into KY-99M equipment Reasoning: There are a lot of components, considerations and possible troubleshooting that goes into, first, setting up the equipment and, second, loading the keymat from the SKL to the KY-99M device to be able to operate the device properly. By separating the PO into those two learning chunks, the board felt it would be a more efficient way to teach while balancing theory and practice.	
4.	The board debated whether or not to have a standards for building a profile load when preparing the SKL to load the keymat, because it is something the members lack knowledge on that may be required to do at their operational unit. However, the board decided not to create a distinctive standard between the two, but have it as a teaching point, because the procedures are specifically laid out in the KY-99M SOP and to prevent a higher chance of operator errors.	
5.	The board agreed that it would be time consuming and logistically difficult to physically inject faults and errors into	

Item	Details	Action
	the KY-99M practical run to make trainees troubleshoot faults. Plus, during teaching some trainees are most likely going to make mistakes where the instructor would have the opportunity to explain and teach some troubleshooting possibilities. Therefore, the idea to have some scenario based written questions to simulate faults would be a more efficient way to evaluate the trainees' ability to troubleshoot faults. However, discussion were had if this should be done in two separate ECs (one written test and one practical), or simultaneously during the one practical EC.	
6.	Practical Labs are to be found with the ICT instructors, so we can reuse them.	Developers to contact ICT instructors or reference material from ICT to obtain Practical Labs.
6.	The board decided not to put in any troubleshooting points on the PC because if they successfully configure the KY-99M from start to finish it is implied that they troubleshot as required without assistance.	

RECORD OF AMENDMENT & AMPLIFICATION

1. The record of amendment and amplification will be filled in whenever a change to the TP is initiated. This section is used to provide amplification details and provide a history of the changes made. In the requirement for the amendment section, the member inputting the changes will transpose the information from the submitted TP Amendment Form, thus ensuring the QSP has the most relevant information.

Version:	**X.X**
Date:	**Who Initiated Change:**
Requirement for the Amendment:	
Description of Changes:	
Change Tracking number:	
Trg Mgr:	

ANNEX M – ACRONYMS AND ABBREVIATIONS

1. This annex is to be utilized only to identify uncommon abbreviations used within this QSP.

AAR	After Action Report
ADM	Assistant Deputy Minister
ATIS	Aerospace telecommunication and information system
ATT	Actual Training Time
BAT	Base Administration Time
BIS	Base Information Services
BIT	Built-in Test
CAF	Canadian Armed Forces
CANFLTPAC	Canadian Fleet Pacific
CCT	Circuit
CF	Canadian Forces
CFAT	Canadian Forces Aptitude Test
CFB	Canadian Forces Base
CFCD	Canadian Forces Communication Doctrine
CFITES	Canadian Forces Individual Training and Education System
CIK	Cryptographic ignition key
CIS	Communication Information Systems
CISN	Communication Information System Network
CISO	Communication Information System Officer
CNPTG	Canadian Naval Personnel Training Group
CNS	Chief of Naval Staff
COA	Course Of Action
COC	Chain of Command
COMM	Communication
COMSEC	Communication Security
CSQ	Common Specialty Qualification
CSS	Communications Security Standard
CT	Cypher Text
DAOD	Defence Administrative Orders and Directives
DIF	Difficulty Intensity Frequency
DND	Department of National Defence
DNIW	Director Naval Information Warfare
DPGR	Director Personnel Generation Requirements

DSP	Display
EC	Enabling Check
ECU	End Cryptographic Unit
ECUI	End Cryptographic Unit Interface
EDI	Explain Demonstrate Imitate
EO	Enabling Objective
ER	Enhanced Reliability
FMF	Fleet Maintenance Facility
GUI	Graphic User Interface
HF	High Frequency
HMCS	His Majesty's Canadian Ship
HOD	Head of Department
HRMS	Human Resources Management Systems
IAW	In Accordance With
ICT	Initial Cadre Training
IM	Information Management
INFOSEC	Information Security
IOT	In order to
IP	Internet Protocol
IS	Information Systems
IT	Information technology
IWS	Information Warfare System
KEK	Key Encryption Key
LCD	Liquid Crystal Display
MESCR	Military Employement Structure Change Request
MESIP	Military Employment Structure Implementation Plan
MITE ID	Military Individual Training and Education Identification
MOSID	Military Occupational Structure Identification Code
MS	Master Sailor
NATO	North Atlantic Treaty Organization
NAV	Naval
NDID	National Defence Identification
NFS	Naval Fleet School
NIC	Network Interface Card
NPTORD	Naval Personel Training Orders
NSN	NATO Stock Number
OCMI	Operational Configuration Managemetn Interface
OFL	Offline
OFP	Operationally Functional Point

OJPR	On the Job Performance Records
OP	Operation
OPI	Office of Primary Interest
OPS	Operations
OTAD	Over the Air Distribution
OTAR	Over the Air Re-keying
OTAT	Over the Air Transfer
PARA	Paragraph
PC	Performance Check
PCR	Performance Check Record
PD	Professional Development
PNL	Panel
POL	Petrolium Oil Lubricant
PP&S	Pen, Paper and Stationary
PPT	Power Point
PT	Plain Text
PTT	Push To Talk
QS	Qualification Standards
QSMS	Qualification Standards Management System
QTY	Quantity
RASKL	Really Simple Key Loader
RCN	Royal Canadian Navy
REF	Reference
RQ	Rank Qualification
SER	Serial
SIG	Signal
SKL	Simple Key Loader
SME	Subject Matter Experts
SOP	Standard Operational Procedures
SWE	Salary Wage Envelop
TACLANE	Tactical Local Area Network Encryptor
TAT	Transport Administration Time
TD	Temporary Duty
TDO	Training Development Officer
TE	Training Establishment
TECH	Technician
TEK	Traffic Encryption Key
TEL	Technology Enabled Learning
TM	Training Manager

TP	Training Plan
TPWB	Training Plan Writing Board
TRB	Training Review Board
TT	Training Technology
TTT	Trianing Travel Time
UHF	Ultra High Frequency
US	Unserviceable
USS	Unit Security Supervisor
VOL	Volume